

MedAxis × Meridian — FHIR-Style Data Dictionary

This data dictionary describes all source tables provided by Meridian Health Group's EPR system. Column definitions, data types, allowed values, and known quality issues are documented below. This is your primary reference for Phase 2 (data warehouse) and Phase 3 (feature engineering).

patients.csv

Column	Type	Description	Known Issues
PATIENT_ID	STRING	Unique patient identifier — format PAT#####	~12 near-duplicate rows with spelling variation
DATE_OF_BIRTH	DATE	Patient date of birth (YYYY-MM-DD)	~3% null — data entry gap pre-2022
GENDER	STRING	M / F / Unknown	~1% null — system migration artefact
ETHNICITY	STRING	Ethnic group — 16-category ONS classification	Inconsistent: 'White-British' vs 'White British'
DEPRIVATION_DECILE	INT	IMD decile 1–10 (1=most deprived)	No known issues
PRIMARY_SITE	STRING	Primary treating hospital (MER001–MER006)	No known issues
REGISTRATION_DATE	DATE	Date of first registration in system	No known issues
POSTCODE_SECTOR	STRING	Outward code + sector (e.g. SW1 9)	Not precise enough for geocoding

admissions.csv

Column	Type	Description	Known Issues
ENCOUNTER_ID	STRING	Unique encounter ID — format ENC#####	~8 duplicate rows — data export error
PATIENT_ID	STRING	Foreign key → patients.PATIENT_ID	No known issues
ADMISSION_DATE	DATE	Date of admission (YYYY-MM-DD)	~0.5% impossible (after discharge date)
DISCHARGE_DATE	DATE	Date of discharge (YYYY-MM-DD)	~1.5% null (still admitted or error)
ADM_TYPE	STRING	EM/EL/TR/MAT — see glossary	No known issues
DISCH_DEST	STRING	Discharge destination — see glossary	No known issues
ICD10_PRI	STRING	Primary ICD-10 diagnosis code	No known issues
ICD10_SEC	STRING	Secondary codes, pipe-separated	Not present pre-2022 for some sites
LOS_DAYS	INT	Pre-calculated length of stay in days	May disagree with date arithmetic — use dates
DD_FLAG	INT	Delayed discharge: 1=delayed, 0=not	No known issues
COMORB_CNT	INT	Count of documented comorbidities	0 may mean not documented, not zero comorbidities

Column	Type	Description	Known Issues
PREV_ADM_90	INT	Admissions in prior 90 days	NULL for records before March 2023
30DR	INT	30-day readmission: 1=readmitted, 0=not	Verify against date arithmetic for validation
WARD_CODE	STRING	Admitting ward — see hospital reference	No known issues
SITE_ID	STRING	Hospital site code — see hospital reference	No known issues

lab_results.csv

Column	Type	Description	Known Issues
LAB_ID	STRING	Unique lab result ID	No known issues
ENCOUNTER_ID	STRING	FK → admissions.ENCOUNTER_ID	No known issues
PATIENT_ID	STRING	FK → patients.PATIENT_ID	No known issues
TEST_NAME	STRING	Lab test name (HBA1C, CREATININE etc.)	No known issues
RESULT_VALUE	FLOAT	Numeric result — see unit	~2% null — machine error
UNIT	STRING	Measurement unit	~1% lowercase inconsistency
LAB_FLAG	INT	0=normal, 1=mildly abnormal, 2=critically abnormal	No known issues
RESULT_DATE	DATE	Date of result	No known issues

weekly_demand_forecast.csv

Column	Type	Description	Known Issues
WEEK_START_DATE	DATE	Week commencing date	~8 rows in DD/MM/YYYY format — inconsistency
WEEK_NUMBER	INT	Sequential week number (1–104)	No known issues
ISO_WEEK	INT	ISO calendar week number (1–52)	No known issues
SITE_ID	STRING	ALL = network aggregate	No known issues
EM_ADMISSIONS	INT	Emergency admissions that week	Week 68: PARTIAL_DATA — pipeline failure
ICU_DEMAND	INT	ICU bed demand	No known issues
BED_OCCUPANCY_PCT	FLOAT	Network-wide bed occupancy %	No known issues
DISCHARGE_VOLUME	INT	Total discharges that week	No known issues
DATA_QUALITY_FLAG	STRING	PARTIAL_DATA if known quality issue	Check before modelling

■■ **CRITICAL: Week 68 in weekly_demand_forecast.csv has DATA_QUALITY_FLAG = PARTIAL_DATA. Emergency admissions for that week are approximately 69% lower than expected. This is a data pipeline failure — not a real demand drop. Candidates who do not identify and handle this will produce an incorrect forecast model.**